

An Approach to Detect Kyphosis Disease with XGBoost in Comparison with Logistic Regression.

INTRODUCTION

- The aim of this study is to compare the performance of XGBoost and logistic regression in detecting kyphosis disease.
- Understanding the performance of different algorithms in detecting kyphosis disease can help improve diagnosis accuracy and treatment outcomes.
- The findings of this study can be applied in the field of healthcare to assist medical professionals in identifying and treating kyphosis disease.
- XGBoost is a powerful machine learning algorithm known for its ability to handle complex relationships and handle missing data effectively. Logistic regression is a popular algorithm used for binary classification tasks, providing interpretable results.
- The dataset used in this study includes variables such as age, gender, start, and kyphosis. These variables are used to train and evaluate the performance of both the XGBoost and Random Forest algorithms in predicting kyphosis disease.

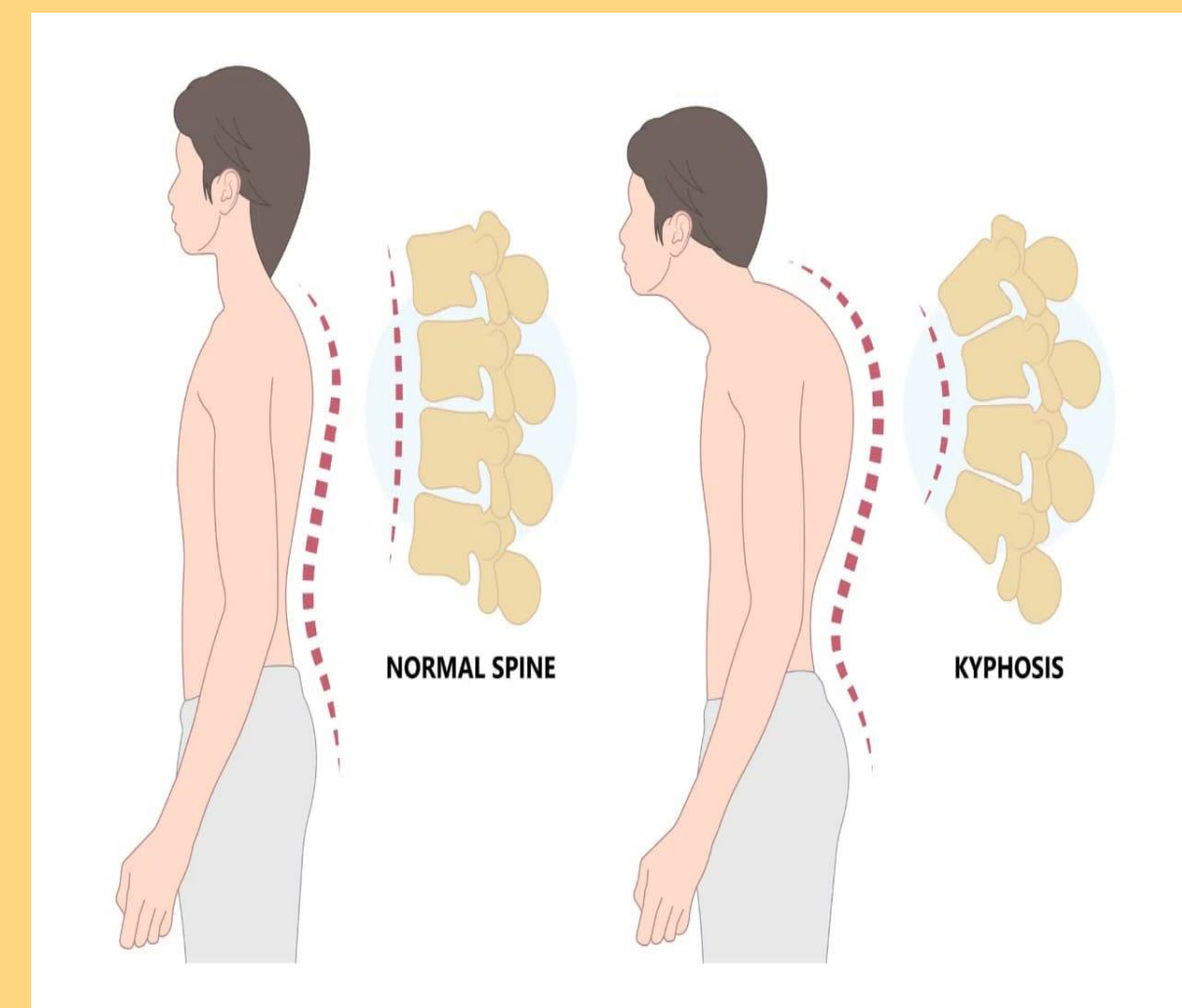
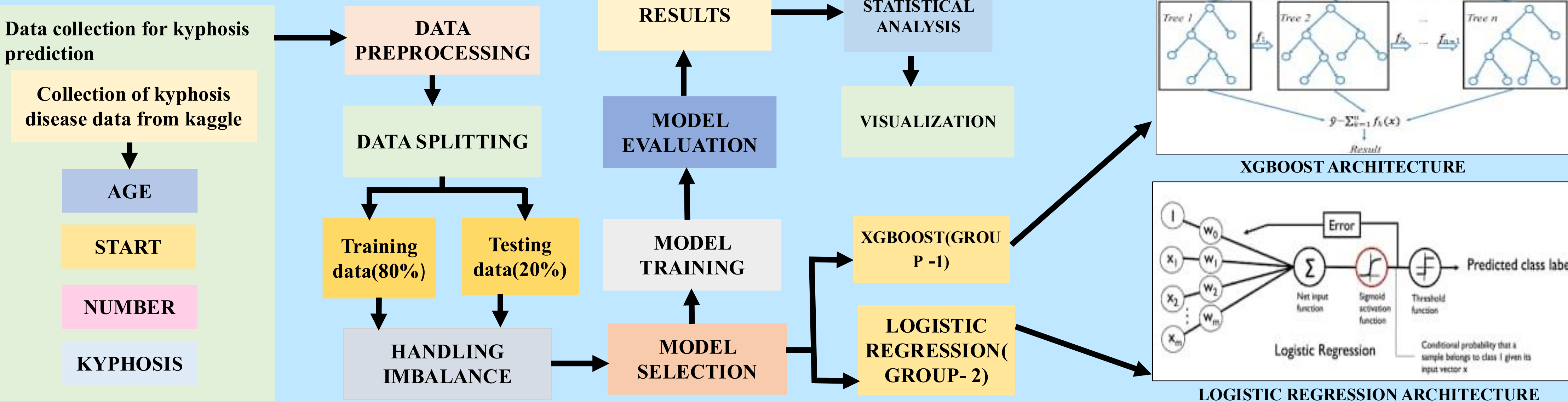


Fig. 1:Normal and affected spine by kyphosis.

MATERIALS AND METHODS



RESULTS

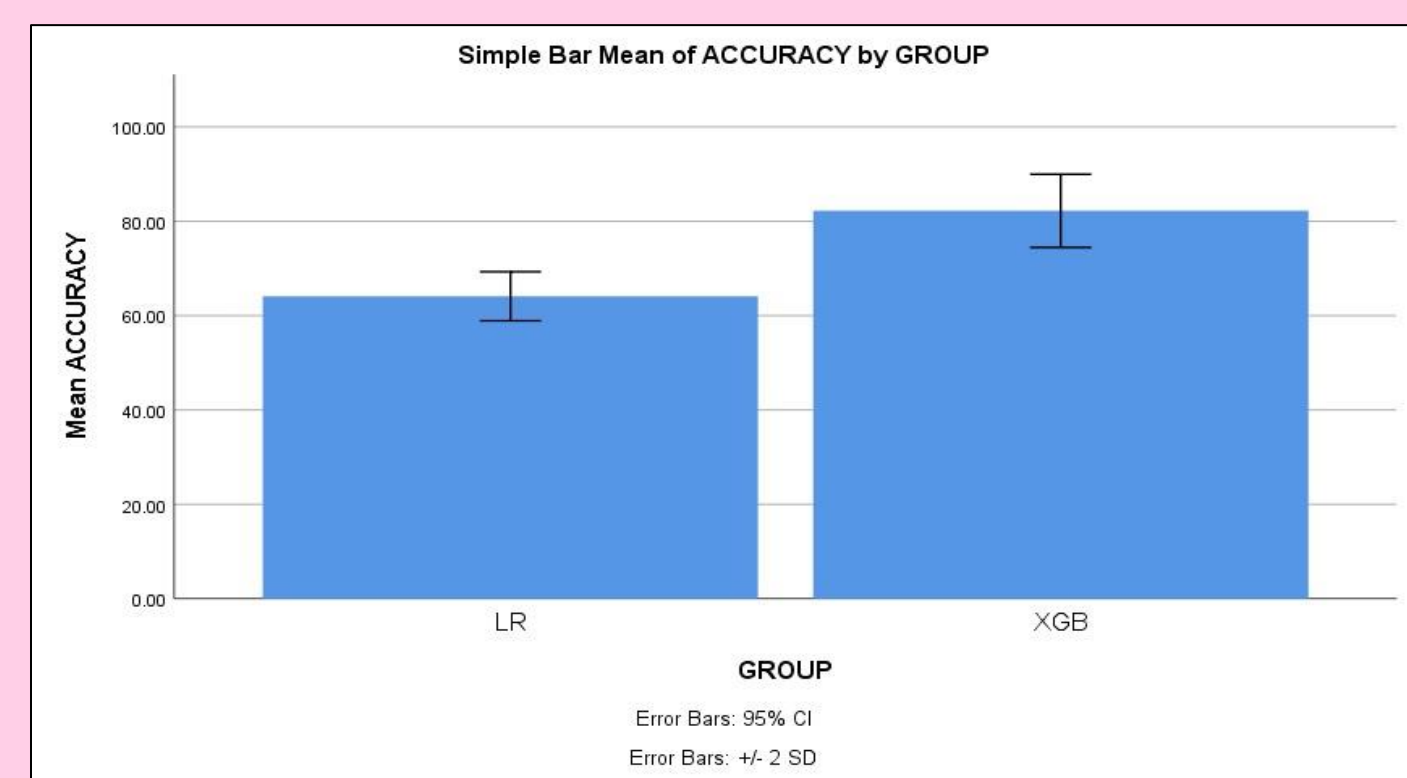


Fig. 2:Comparison of kyphosis detection between Xgboost and Logistic regression.

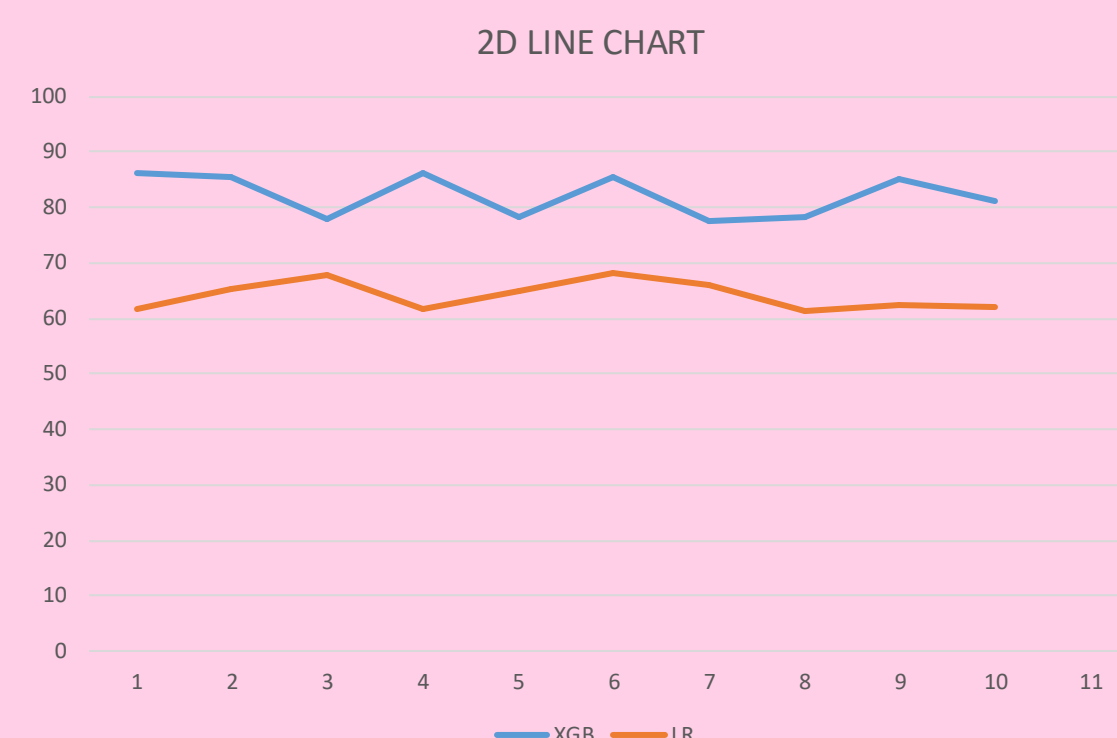


Fig. 3:Comparison of accuracy values of kyphosis detection with Xgboost and Logistic regression.

GROUP		N	Mean	Std. Deviation	Std. Error Mean
ACCURACY	XGB	10	82.2140	3.87330	1.22484
	LR	10	64.1030	2.60132	0.82261

Table. 1:Comparison of the mean accuracy of kyphosis detection with Xgboost and Logistic regression

DISCUSSION AND CONCLUSION

- Based on T-test statistical analysis, the significance value of $p=0.001$ (independent sample T-test $p<0.05$) is obtained and shows that there is a statistical significant difference between the group 1 and group 2 (group 1-Xgboost & group 2-Logistic regression).
- The Logistic regression algorithm exhibited an accuracy rate of 64.71%, whereas the XgBoost algorithm, demonstrated an accuracy of 82.35% in predicting kyphosis disease in individuals.
- Further research could focus on optimizing XGBoost's computational efficiency and enhancing its interpretability while improving predictive performance in kyphosis detection.
- Age, gender, and other factors can affect the accuracy of kyphosis prediction using machine learning. However, limitations arise from data availability, quality, potential bias, and the need for validation on diverse populations.
- Machine learning approaches, specifically Xgboost and Logistic regression algorithms, show promise in accurately predicting kyphosis. However, more research and validation are necessary to address limitations and improve the overall effectiveness of these approaches.

BIBLIOGRAPHY

- Dankwa, Stephen, and Wenfeng Zheng. 2019. "Special Issue on Using Machine Learning Algorithms in the Prediction of Kyphosis Disease: A Comparative Study" *Applied Sciences* 9, no. 16: 3322. <https://doi.org/10.3390/app9163322>.
- Lee CH, Jo DJ, Oh JK, Hyun SJ, Park JH, Kim KH, Bae JS, Moon BJ, Lee CK, Shin MH, Jang HJ, Han MS, Kim CH, Chung CK, Moon SM. Development and Validation of an Online Calculator to Predict Proximal Junctional Kyphosis After Adult Spinal Deformity Surgery Using Machine Learning. *Neurospine*. 2023 Dec;20(4):1272-1280. doi: 10.14245/ns.2342434.217. Epub 2023 Dec 31. PMID: 38171294; PMCID: PMC10762414.
- Chauhan, A.S.; Lilhore, U.K.; Gupta, A.K.; Manoharan, P.; Garg, R.R.; Hajje, F.; Keshta, I.; Raahemifar, K. Comparative Analysis of Supervised Machine and Deep Learning Algorithms for Kyphosis Disease Detection. *Appl. Sci.* 2023, 13, 5012. <https://doi.org/10.3390/app13085012>.
- Hussein, Yehia Y., and Muhammad Mohsin Khan. "Using Artificial Intelligence to Predict the Development of Kyphosis Disease: A Systematic Review." *Cureus* 15, no. 11 (2023).doi:10.7759/cureus.48341.